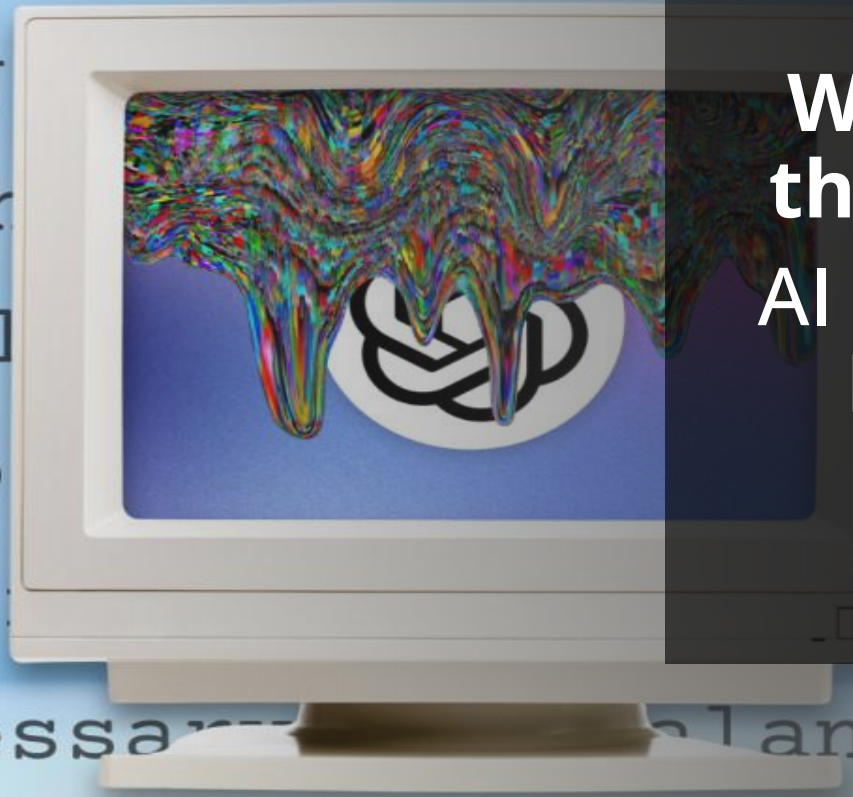


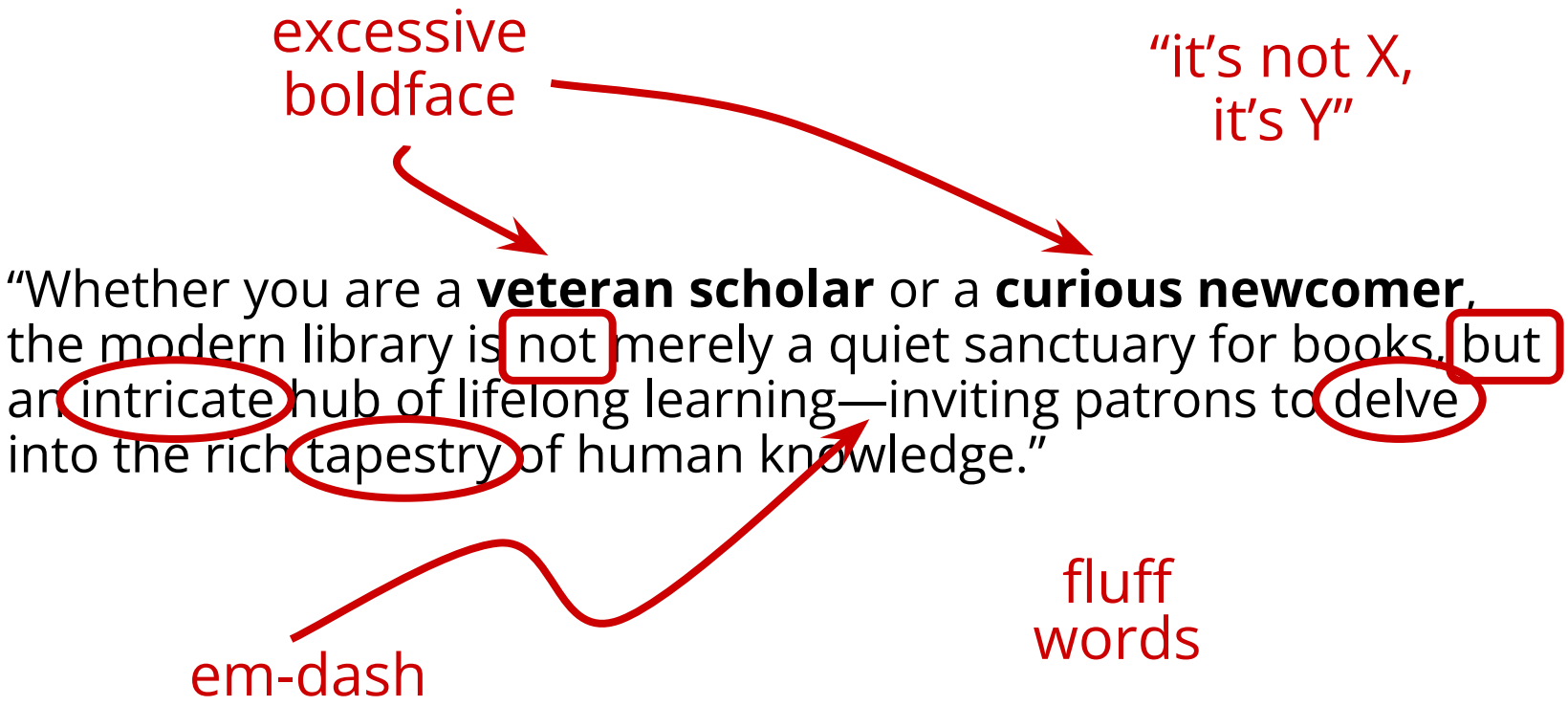
Words From the Machine:

AI and Human Language

Thomas Juzek
Zina Ward

FSU

© Science Friday


excessive boldface
“it’s not X, it’s Y”
“Whether you are a **veteran scholar** or a **curious newcomer**, the modern library is not merely a quiet sanctuary for books, but an intricate hub of lifelong learning—inviting patrons to delve into the rich tapestry of human knowledge.”
em-dash
fluff words

LLMs use language in distinctive ways.

"Whether you are a **seasoned scholar** or a **curious newcomer**, the modern library is not merely a quiet sanctuary for books, but an intricate hub of lifelong learning—inviting patrons to delve into the rich tapestry of human knowledge."

How? Why? And what's it doing to how *we* use language?

word choice/use

em-dashes

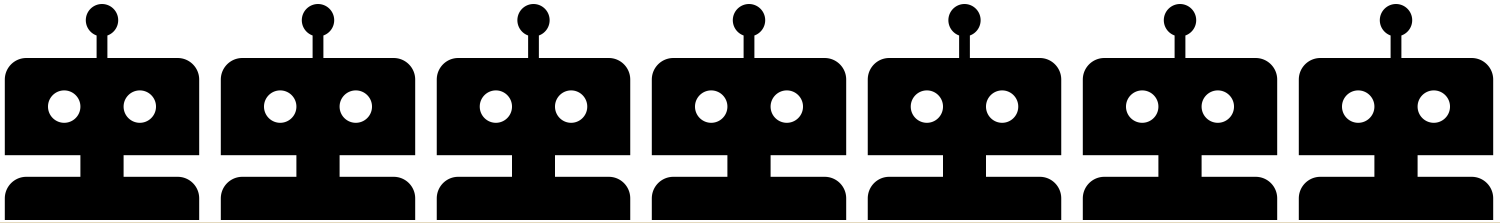
- lexical selection
- register/tone
- syntactic structures
- sentence length
- punctuation/formatting
- information density
- voice
- …

How? Why? And what's it doing to how *we* use language?

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What words do LLMs (over)use? Why? PART 1

How is AI shaping human language? PART 2



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Why Does ChatGPT "Delve" So Much? Exploring the Sources of Lexical Overrepresentation in Large Language Models

Tom S. Juzek and Zina B. Ward*
Florida State University
tjuzek@fsu.edu, zward@fsu.edu

Abstract

Scientific English is currently undergoing rapid change, with words like "delve," "intricate," and "underscore" appearing far more frequently than just a few years ago. It is widely assumed that scientists' use of large language models (LLMs) is responsible for such trends. We develop a formal, transferable method to characterize these linguistic changes. Application of our method yields 21 focal words whose increased occurrence in scientific abstracts is likely the result of LLM usage. We then pose "the puzzle of lexical overrepresentation": *why* are such words overused by LLMs? We fail to find evidence that lexical overrepresentation is caused by model architecture, algorithm choices, or training data. To assess whether

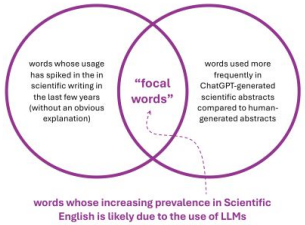


Figure 1: We formalize a procedure for identifying words whose increasing prevalence is likely the result of LLM usage. Although our focus is Scientific English, the method can be applied across domains and

cs.CL] 3 Aug 2025

Word Overuse and Alignment in Large Language Models: The Influence of Learning from Human Feedback

Tom S. Juzek[1][0000−0002−3204−3879] and Zina B. Ward[1][0000−0003−0160−6656]*

Florida State University, Tallahassee FL 32306, USA

Abstract. Large Language Models (LLMs) are known to overuse certain terms like "delve" and "intricate." The exact reasons for these lexical choices, however, have been unclear. Using Meta's Llama model, this study investigates the contribution of Learning from Human Feedback (LHF), under which we subsume Reinforcement Learning from Human Feedback and Direct Preference Optimization. We present a straightforward procedure for detecting the lexical preferences of LLMs that are potentially LHF-induced. Next, we more conclusively link LHF to lexical overuse by experimentally emulating the LHF procedure and demonstrating that participants systematically prefer text variants that include certain words. This lexical overuse can be seen as a sort of misalignment, though our study highlights the potential divergence between the lexical expectations of different populations – namely LHF workers versus LLM

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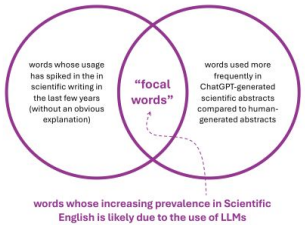


Figure 1: We formalize a procedure for identifying words whose increasing prevalence is likely the result of LLM usage. Although our focus is Scientific English, the method can be applied across domains and

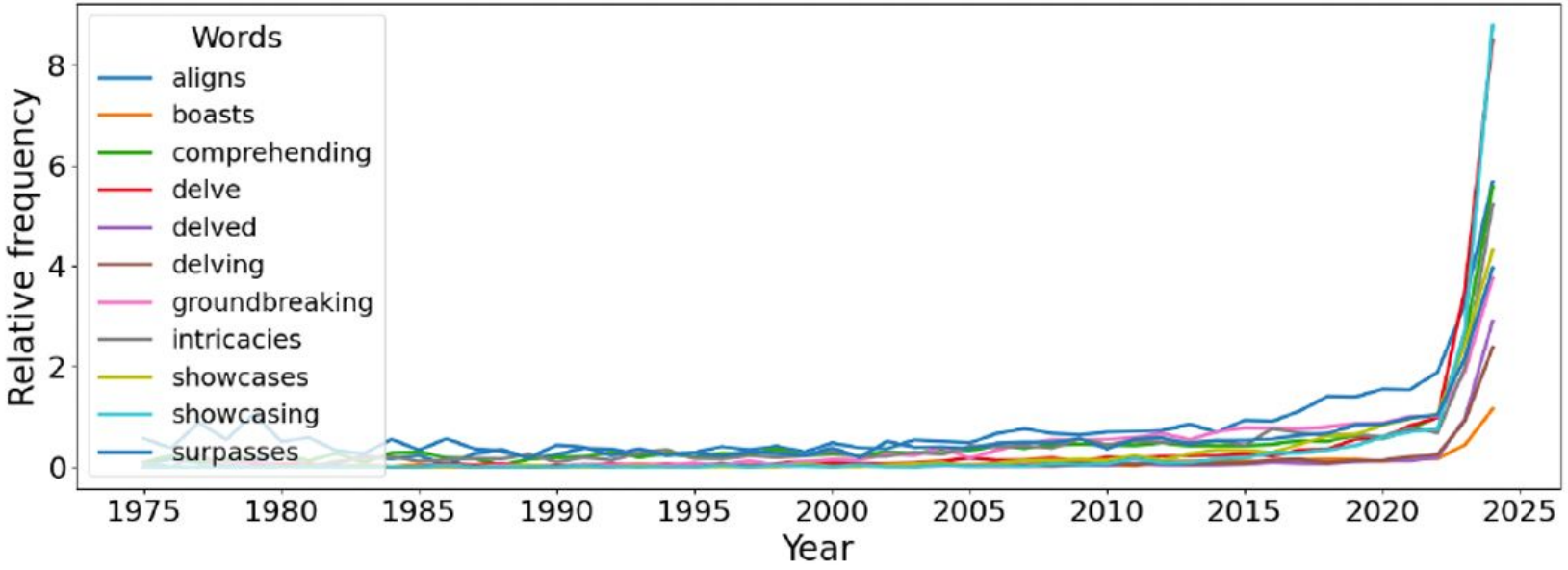
(An) Aim: Identify words increasingly appearing in scientific English whose recent usage spike is likely to due to the advent of LLMs

Word	opm 2020	opm 2024	Incr. %
delves	0.21	14.38	6697.14
delved	0.12	2.90	2240.47
delving	0.12	2.38	1816.83
showcasing	0.59	8.79	1396.03
delve	0.58	8.50	1374.92
boasts	0.11	1.15	918.18
underscores	4.50	45.19	903.61
comprehending	0.56	5.58	898.95
intricacies	0.60	5.22	772.85
surpassing	1.37	10.50	667.48
intricate	6.22	44.22	611.24
underscoring	2.70	17.17	536.94
garnered	2.44	13.13	437.19
showcases	0.82	4.31	422.45
emphasizing	8.30	41.27	397.12
underscore	7.42	36.40	390.65
realm	2.25	10.85	381.10
surpasses	0.85	3.96	367.55
groundbreaking	0.87	3.75	330.42
advancements	12.49	47.17	277.59
aligns	1.55	5.68	266.97

words whose recent spike in usage ***in Scientific English*** is likely due to to the use of LLMs

words used more by AI than human writers in ***news text***

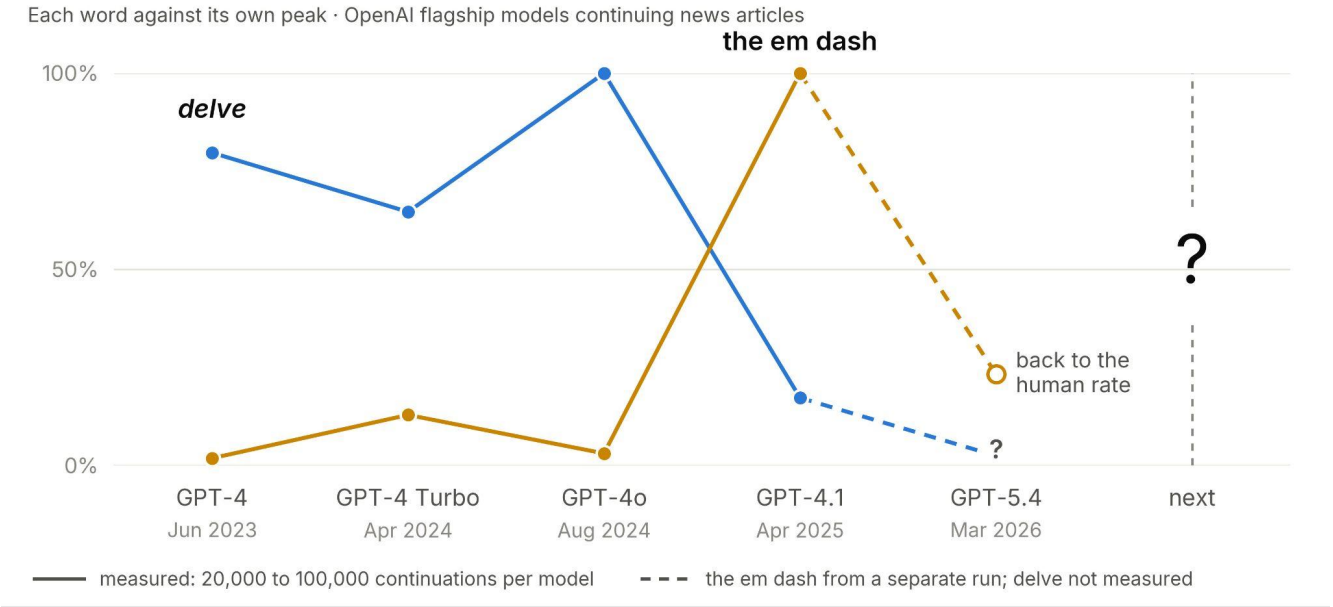
#	Lemma	POS	c_H	c_M	LPR
1	additionally	ADV	0	421	6.74
2	emphasize	VERB	16	5 180	5.75
3	revolutionize	VERB	0	59	4.78
4	revitalize	VERB	0	44	4.49
5	captivated	VERB	1	94	4.14
6	conservationist	NOUN	0	31	4.14
7	streamlin[e]	VERB	0	27	4.01
8	firsthand	ADV	0	25	3.93
9	personalize	VERB	1	74	3.91
10	introspective	ADJ	0	24	3.89


Words
aligns
boasts
comprehending
delve
delved
delving
groundbreaking
intricacies
showcases
showcasing
surpasses
Relative frequency
0
2
4
6
8
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year

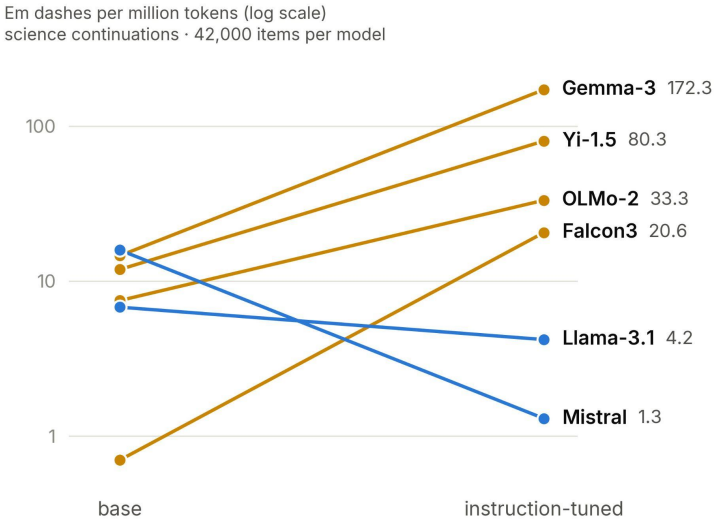
Similar words across languages

Concept	n	Languages and lemmas
EMPHASISE / STRESS / HIGHLIGHT (VERB)	24/34	AF: *beklemtoon*, AR: أَشَّر, BG: акцентирам, подчертавам, подчертая, CS: *zdůraznit, zdůrazňovat, podtrhovat, vyzdvihnout*, DE: *betonen, hervorheben*, EL: υπογραμμίζω, EN: *emphasize, highlight*, ES: *enfatizar, destacar, subrayar, realzar*, ET: *rõhutama*, FI: *korostaa, korostua*, FR: *insister, marquer*, HR: *naglasiti*, ID: *tekan, sorot*, IT: *sottolineare, evidenziare*, LT: *pabrėžti*, LV: *uzsvērt*, NL: *benadrukken*, PL: *podkreślać, podkreślić*, PT: *enfatizar, destacar, ressaltar*, RO: *sublinia, evidenția*, RU: подчеркивать, TR: *vurgula*, UK: підкреслювати, підкреслити, наголошувати, ZH: 强调
IMPORTANCE / SIGNIFICANCE / PRIORITY (NOUN)	20/34	AR: أَهَمِّيَّة, BG: важност, CS: *důležitost, důraz*, DE: *bedeutung, notwendigkeit, dringlichkeit*, EL: σημασία, EN: *importance*, ES: *importancia*, ET: *olulisus, võtme_tähtsus*, FA: اولویتبندی, FR: *importance, nécessité*, HI: प्राथमिकता, HR: *važnost*, ID: *penting(nya)*, IT: *importanza*, PL: *priorytet, znaczenie*, PT: *importância*, RO: *importanță*, RU: важность, SR: *važnost*, UK: важливість
INNOVATIVE / GROUNDBREAKING (ADJ)	18/34	CS: *inovativní, inovace, inovací, moderní*, DE: *innovativ, modern*, EN: *innovative, groundbreaking, advanced*, ES: *innovador*, ET: *innovaatiline*, FI: *innovatiivinen*, FR: *innovant*, ID: *inovatif*, IT: *innovativo*, KK: инновациялық, LT: *inovatyvus, modernus*, NL: *innovatief*, PL: *innowacyjny, nowoczesny, zaawansowany*, PT: *inovador*, RO: *inovator, revoluționar*, RU: инновационный, UK: інноваційний, ZH: 先进

LLMs’ favorite constructions change over time



Model-to-model variation



PNAS RESEARCH ARTICLE | COMPUTER SCIENCES

Do LLMs write like humans? Variation in grammatical and rhetorical styles

Alex Reinhart[a,1], Ben Markey[b], Michael Laudenbach[c], Kachatad Pantusen[a,d], Ronald Yurko[a], Gordon Weinberg[a], and David West Brown[b]

Edited by Jeffrey Ullman, Stanford University, Stanford, CA; received November 5, 2024; accepted January 7, 2025

Large language models (LLMs) are capable of writing grammatical text that follows instructions, answers questions, and solves problems. As they have advanced, it has become difficult to distinguish their output from human-written text. While past research has found some differences in features such as word choice and punctuation and developed classifiers to detect LLM output, none has studied the rhetorical styles of LLMs. Using several variants of Llama 3 and GPT-4o, we construct two parallel corpora of human- and LLM-written texts from common prompts. Using Douglas Biber's set of lexical, grammatical, and rhetorical features, we identify systematic differences between LLMs and humans and between different LLMs. These differences persist when moving from smaller models to larger ones and are larger for instruction-tuned models than base models. This observation of differences demonstrates that despite their advanced abilities, LLMs struggle to match human stylistic variation. Attention to more advanced linguistic features can hence detect patterns in their behavior not previously recognized.

Significance

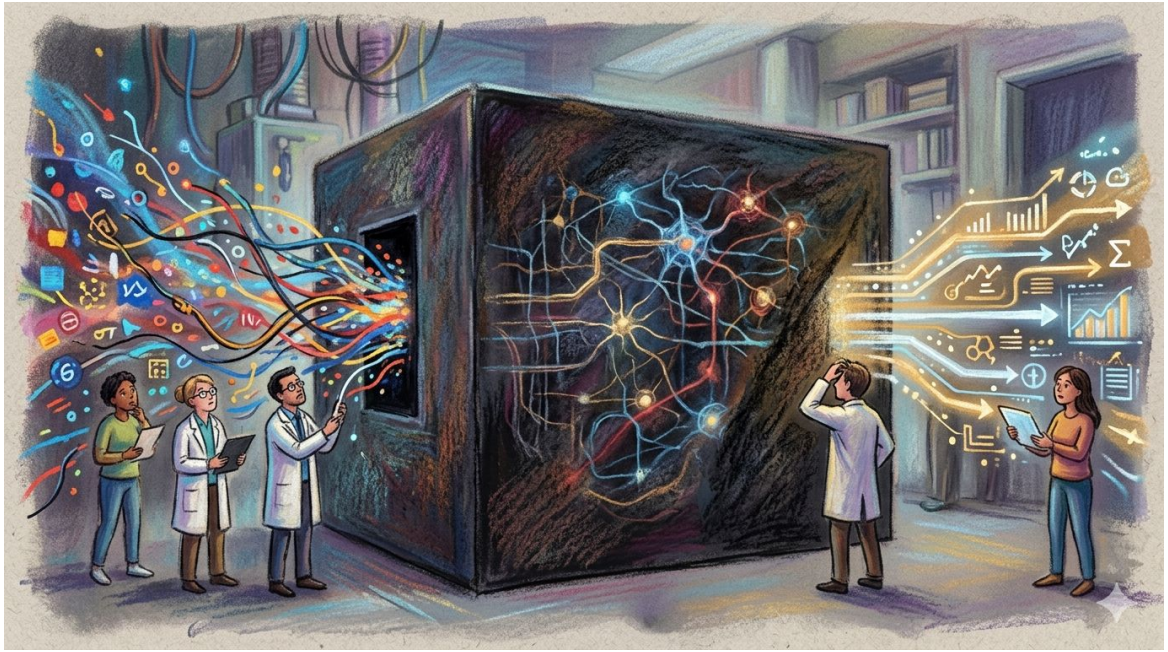
As large language models (LLMs) have grown in power and become more widely available, research has focused on their ability to complete various tasks and the biases they exhibit when doing so. In this study, we instead examine their writing style in detail. We show that instruction-tuned models, which are trained to answer questions and solve

Why are certain words used so much more often by LLMs?



	2020	2024	
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groundbreaking	0.87	3.75	33
vancements	12.49	47.17	
	1.55		

LLMs: The Black Box Problem

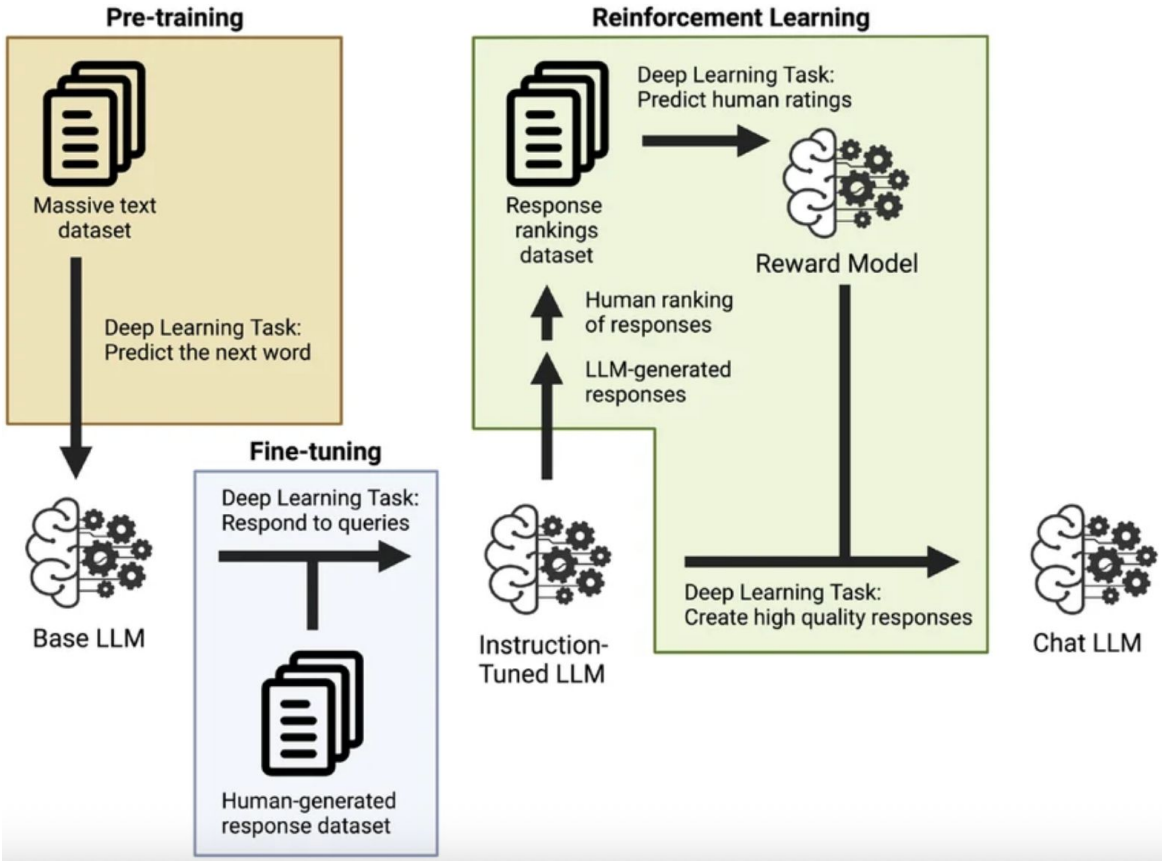




Lexical Overrepresentation in LLMs: Possible Explanations

1. Initial Training Data
2. Fine-Tuning Training Data
3. Model Architecture
4. Choice of Algorithms
5. Context Priming
6. Learning from Human Feedback (LHF)
7. Other Factors

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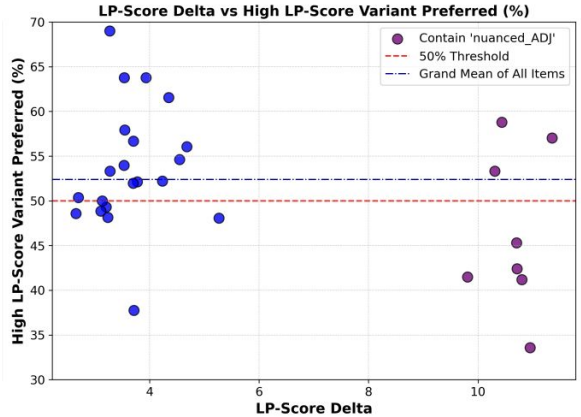
Hypothesis: Lexical overuse is attributable (in part) to learning from human feedback.

Human evaluators exhibit a preference for certain words. LLMs learn, propagate, amplify that preference.

https://www.researchgate.net/figure/Training-a-chat-based-LLM-Training-a-chat-based-LLM-requires-a-multi-stage-learning_fig2_378394229

Experimental test of the LHF hypothesis

We showed participants one abstract containing a lot of LLMs' favorite words, and another abstract containing few. Would they show a preference for abstracts *with* those words?



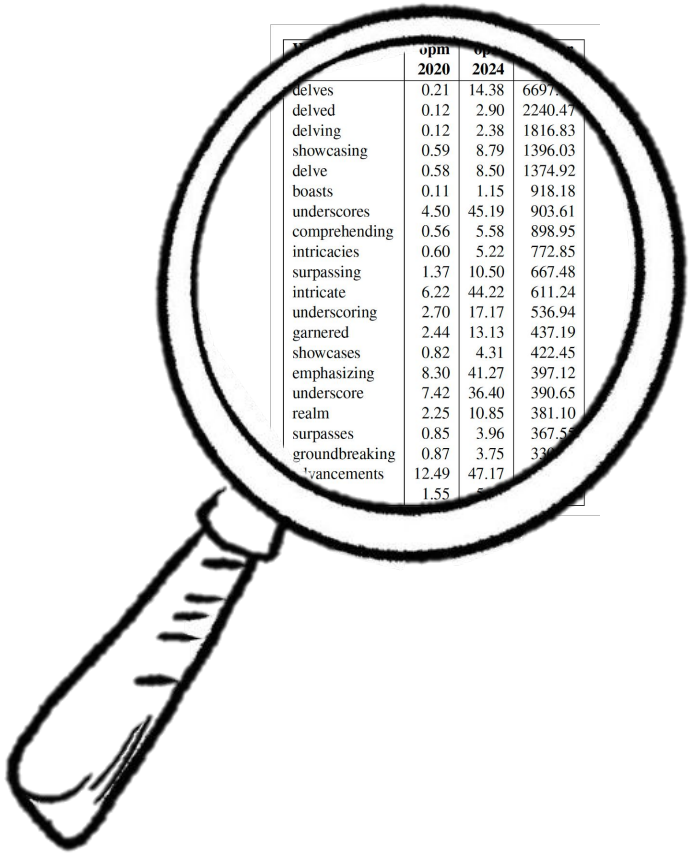
(YES)

*though not for "delve" or (later) em-dashes; stay tuned…

Why are certain words used so much more often by LLMs?

In part and in some cases, because human evaluators exhibit a *preference* for these words…

…and that preference is reinforced during model training.



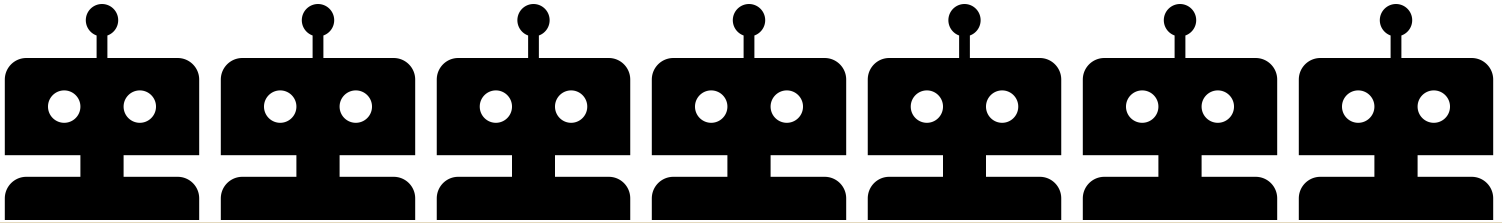
(But WHY do human evaluators prefer these words??)



- because they are under time pressure, and these words look good when one skims (i.e., words are acting as proxy for quality)?
- because human evaluators tend to come from certain demographic groups (e.g., younger, from particular regions) in which such preferences are more common?
- because these words seem to signal sophistication across a broad range of contexts, so they receive high ratings in a variety of feedback tasks?

What words do LLMs (over)use? Why? **PART 1**

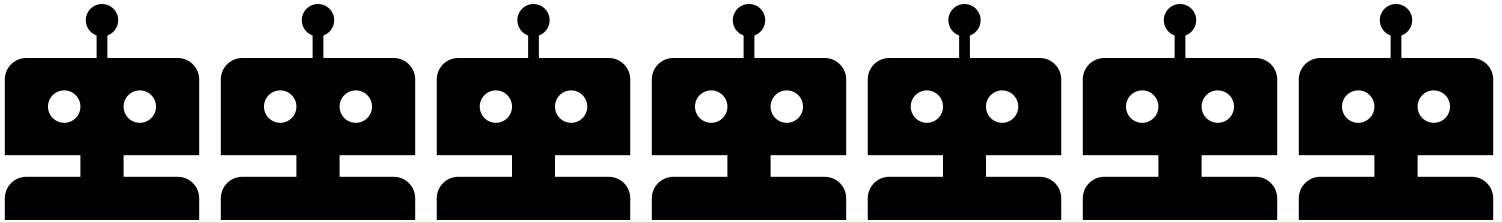
How is AI shaping human language? PART 2



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What words do LLMs (over)use? Why? PART 1

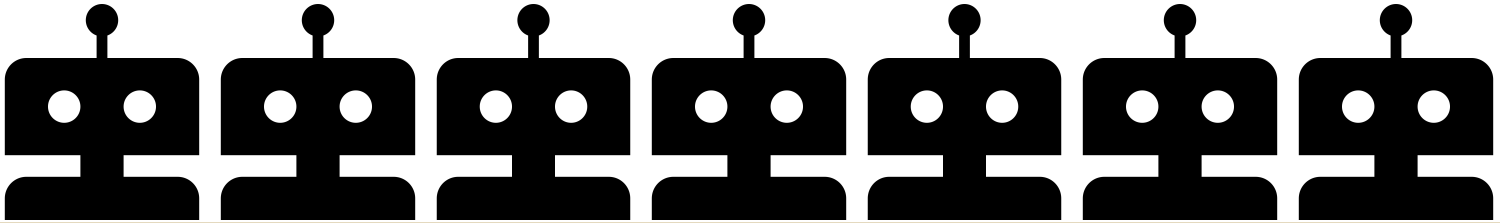
How is AI shaping human language? **PART 2**



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What words do LLMs (over)use? Why? PART 1

“Shaping”, embed in wider discourse PART 2



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How AI is shaping human language

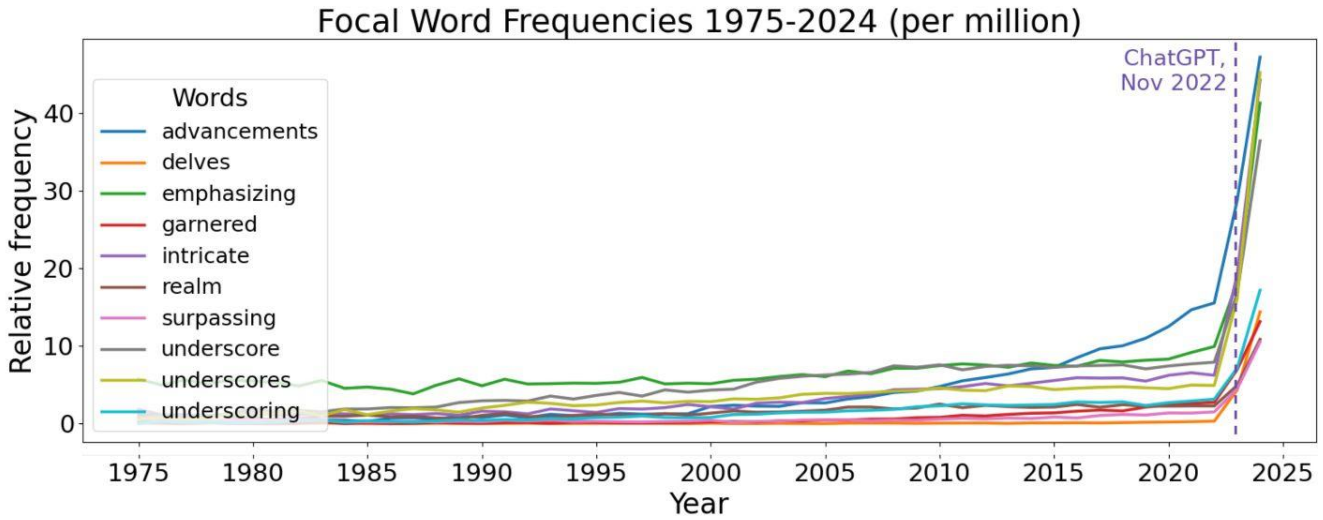
"Shaping" can mean various things

A · Production	A disruption in *produced* language: AI-typical words spike in written text from 2023.
B · Authorship	AI-produced, AI-assisted or human-produced? Mostly unknown: **authorship indeterminacy**.
C · The wider sense	Written language is undergoing a huge disruption. **1** AI as the cause: very plausible, but hard to show conclusively **2** Human input fed the overuse; many words were already rising **3** The real question: what happens to the human language system? **4** Exposure does something to us: repulsion (*delve*), adoption?

Anderson, Galpin & Juzek, AIES 2025 · Juzek & Ward, COLING 2025 · Matsui, *Perspect. Med. Educ.* 2025

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Already rising before ChatGPT



Several of these words climbed for decades; the 2023 jump came on top.

Juzek & Ward, COLING 2025, Figure 4 (top panel): PubMed abstracts, occurrences per million words, 1975–2024

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How much are we being exposed to AI language?

~ 35%

of dated web pages published since ChatGPT show signs of AI writing (July 2026 crawl)

Pew Research Center 2026; Open Pangram detector

≥ 13.5%

of 2024 PubMed abstracts were processed with LLMs (a lower bound)

Kobak et al. 2025 estimate

~ 30%

of US STEM dissertations filed in 2026 contain AI-generated writing (0% before 2023)

Gross et al. 2026, working paper; Pangram

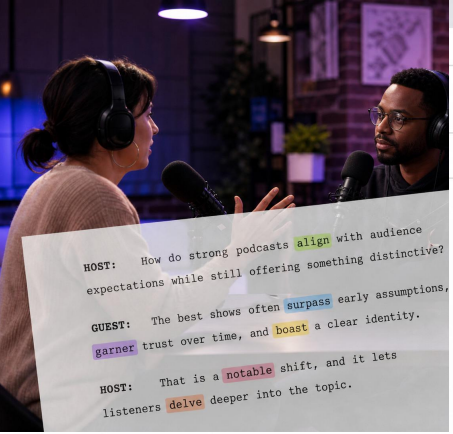
Direct usage data: ~ 50% of US adults use AI chatbots, ~ 25% daily (Pew, 2026).

And useful: ChatGPT Pro proved $p_c < 1$, a percolation question open since 1996; Fields medallist Hugo Duminil-Copin calls the proof “short and particularly elegant”.

Arguably, we are exposed to a lot of AI output these days.

Pew 2026 · Kobak et al., *Sci. Adv.* 2025 · Gross et al., NBER WP 2026 · Duminil-Copin, blog, Aug 2026 (unrefereed)

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alexandra lee
@alexandralee ·

A meticously des
can still feel intric
nuanced, and sur
elegant.

10:42 AM · May 18, 2024 · Twi

City unveils plan to expand green
https://www.dailyreport.com/news/city-unveils-plan-to-
DAILY REPORT
Home Latest World Politics Business
NEWS
City unveils pla green spaces an
The initiative has the potentia crucially supporting greener,



By Emma Collins
Staff Writer
May 15, 20



City officials today unveiled a visionary p urban core, a cornerstone of the city's broader strategy to elevate livability, resilience, and community well-being for generations to come.

Crafted over 18 months with extensive public input, expert guidance, and thoughtful design, the plan will invest $120 million in new parks, restored wetlands, and a rich tree canopy—implemented meticulously to ensure lasting impact. Signature projects feature intricate stormwater systems, vibrant native plantings, and recreational areas for all ages, reflecting the city's nuanced understanding of what people need to thrive in a changing world.

Mayor Daniel Ruiz emphasised the importance of connecting people, protecting nature, and building healthier neighbourhoods. "This plan shows our

Library

Today

Why do cities need more green spaces?
Benefits of remote work
Ideas for sustainable living
How does solar energy work?

Dark mode
Settings
Help & feedback

A Alex

Green spaces offer a multi-faceted benefit for cities: they can significantly improve well-being, reduce heat, and support biodiversity. I suggest expanding parks and tree corridors; additionally, these spaces can strengthen community life.

10:26 AM

Message the assistant...

AI responses can be inaccurate or misleading. Learn more



Who wrote it?

> “I don’t think anyone has reliable information about post-2021 language usage by humans.
>
> Robyn Speer, on freezing the *wordfreq* word-frequency data

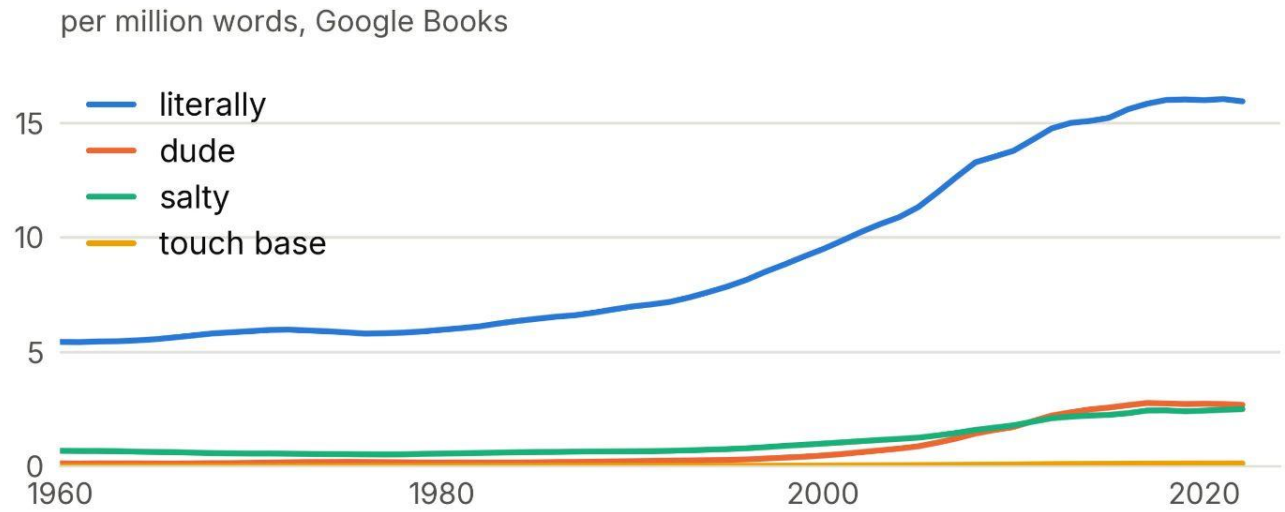
Before LLMs, “the default assumption was that any given text was authored by a human. This presumption no longer holds”: **human-authorship indeterminacy** (Anderson, Galpin & Juzek 2025).

It will complicate “any scientific field that relies on natural language as a proxy for human behavior and cognition” (Anderson et al. 2025).

Speer, github.com/rspeer/wordfreq/blob/master/SUNSET.md, September 2024 · Anderson, Galpin & Juzek, AIES 2025

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Comparison: natural language change

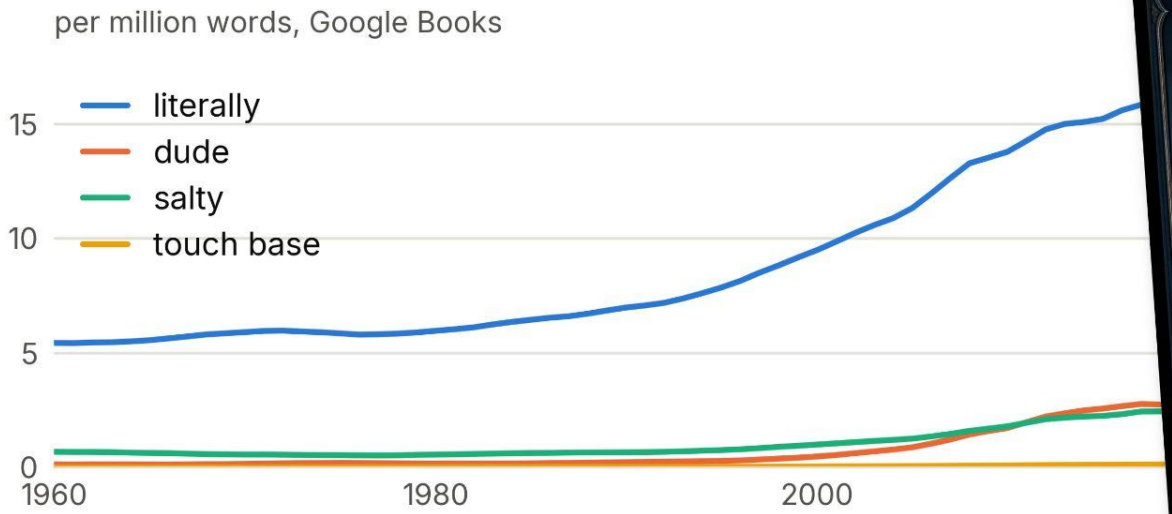


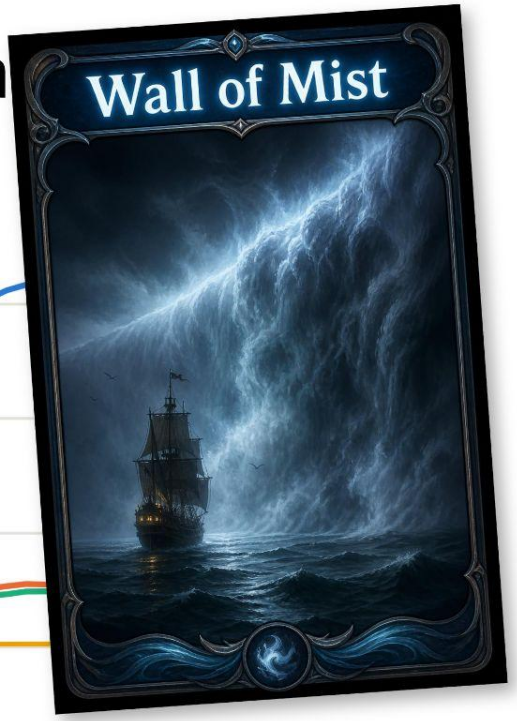
An S-curve over 10–25 years, not a step function like *delve*.

Google Books Ngram Viewer (corpus "en", smoothing 3), 1960–2022, retrieved 19 October 2026

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Comparison: natural language cha





A simple task in the past, “how frequent is X?”, has moved behind a wall of mist.

Google Books Ngram Viewer (corpus “en”, smoothing 3), 1960–2022, retrieved 19 October 2026

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Sidenote: Detection

Genuine, unedited human text	Rarely flagged by Pangram in 2026: 0 of 306 pre-LLM abstracts (Karr et al.); 6 of ~126,000 pre-2023 dissertations (Gross et al.). Caveat: such texts may be in its training data. Non-native English, the 2023 false-positive problem (Liang et al.: 61.3%), is unresolved: 0% on pre-LLM non-native abstracts (Gross et al.), no independent re-test, t.o.k.
AI-touched human text	Flagged by Pangram: 43% after very minor LLM edits (Saha & Feizi); ~64–80% after a full AI rewrite (Karr et al.).
Humanised AI text	Under 4% flagged after a commercial humaniser (Karr et al.); 92.5% after prompt-based humanising (Van Vlasselaer et al.).

Late 2026: arguably credible on untouched text, not yet on edited or humanised.

Karr et al., arXiv 2026 · Gross et al., NBER WP 2026 · Saha & Feizi, ACL Findings 2025 · Van Vlasselaer et al., *IJEI* 2026

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Sidenote: Usefulness

Maths	• $p_c < 1$, open since 1996. AI's proof: "short and particularly elegant" (Duminil-Copin) • IMO gold (2025); $\theta(p_c) = 0$ received a Lean certificate (2026)
Astrophysics	• Goodman's group, stuck for years on spiral-arm motion: ChatGPT solved it in minutes
Coding	• HumanEval: 28.8% (2021) → 92.0% (2024) • SWE-bench Verified: 49.0% → 80.9%, then retired as "effectively saturated"
Music	• Deezer: ~90,000 fully AI-generated tracks uploaded a day (June 2026) • Blind test: 97% could not tell AI music from human music

Duminil-Copin 2026 · Sokol, *Science* 2026 · Chen et al. 2021; OpenAI 2026 · Deezer 2025–26 (vendor figures)

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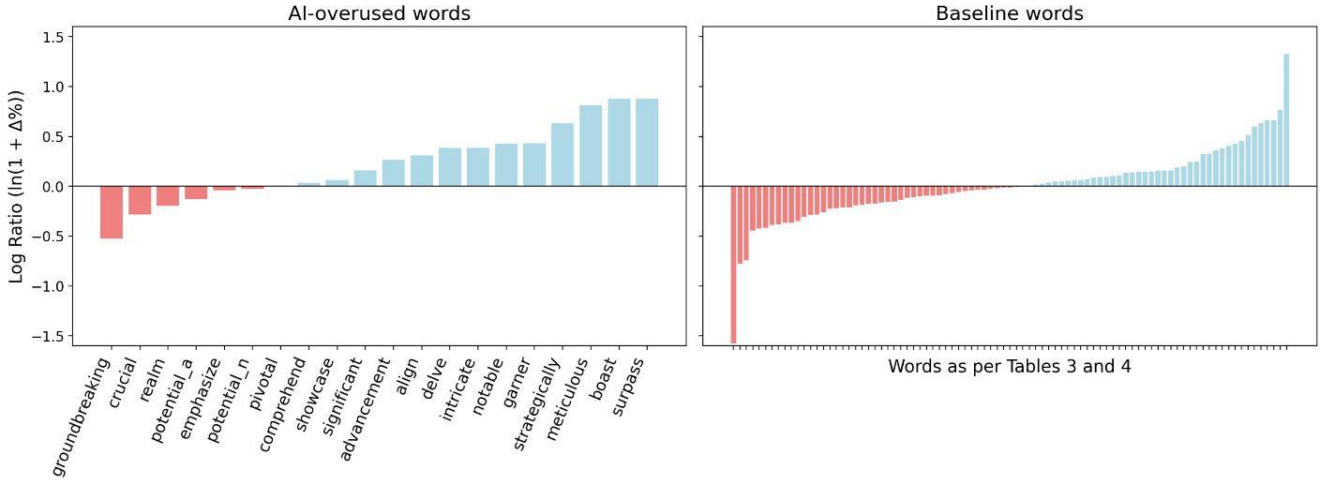
Studying *human* language: unscripted speech

Issue	Post-2022, how can we study the *human* language system (→ authorship indeterminacy)?
Approach	(Semi-)spontaneous speech as a window into the human language system.
Method	• 1,326 science and technology podcast episodes, 22.1M words • 2019–21 vs 2023–25 (2022 left out); each podcast in both • 20 AI-associated words vs 87 synonyms; occurrences per million

Anderson, Galpin & Juzek, "Model Misalignment and Language Change", AIES 2025

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Studying *human* language: results



After 2022, in contrast to their synonyms (weighted log ratio 0.03, n.s.), AI-associated words rose significantly (0.21, $p < 0.001$).

Anderson, Galpin & Juzek, AIES 2025, Figure 2: 22.1 million words of unscripted podcast speech

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What to do with this? Three hypotheses

Almost certainly human speech shows the uptick. What is AI's role?

1 · Nothing to do with it	**The rise is independent of AI.** → There is no counterfactual, so this cannot be ruled out.
2 · Amplifying	**AI speeds up drifts already under way.** → Many of these words were rising before 2022 (shown earlier; Matsui 2025).
3 · Driving	**Exposure to AI causes the change.** → Yakura et al. (preprint) argue a causal link; in a lab study (N = 496) people adopted a chatbot's words after one exchange.

Ideally: a longitudinal study of speakers' AI exposure and language use.

Anderson, Galpin & Juzek, AIES 2025 · Matsui 2025 · Yakura et al., arXiv 2409.01754 v4, 2026 (preprint)

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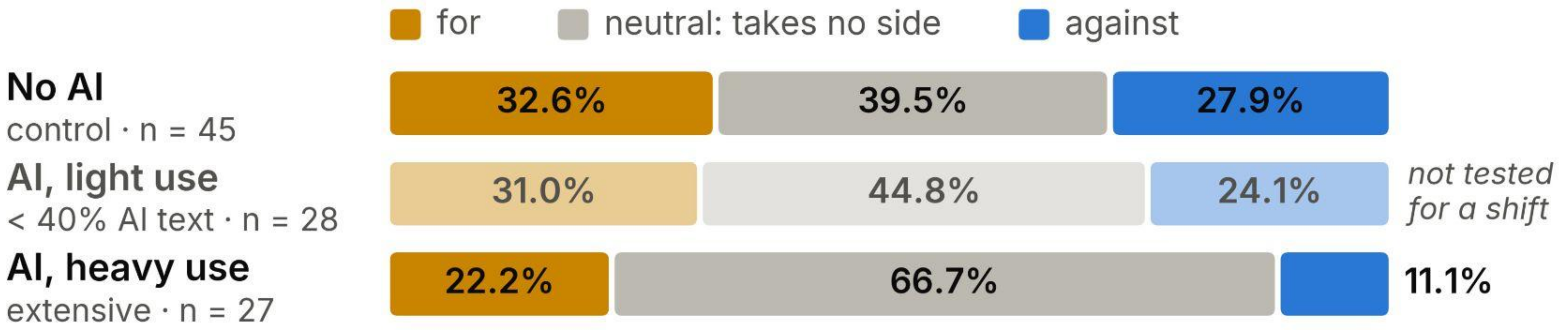
Why care?

> “With everything going on, this is just words, schmords. Who cares?!

Words are the canary

Abdulhai et al. (2026): We think we are doing a grammar check, but AI may shift sentiment (which may be unbeknownst to the user).

Essays on “Does money lead to happiness?”

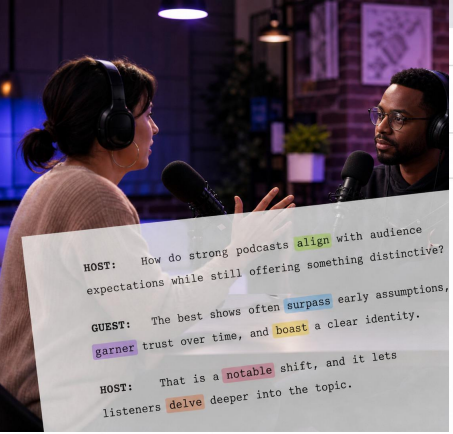


100 US adults (Prolific), randomised: no AI, or an embedded gpt-4o-mini. Stance analysis with GPT-4o.

Heavy use vs control: neutral essays +68.9% (relative; $p < 0.05$). Grammar-only LLM edits of 86 essays from 2021 still shifted their meaning.

Abdulhai, White, … Kleiman-Weiner & Jaques, arXiv 2603.18161 v2, 2026 (preprint): Fig. 6, redrawn

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alexandra lee
@alexandralee ·

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10:42 AM · May 18, 2024 · Twi

City unveils plan to expand green
https://www.dailyreport.com/news/city-unveils-plan-to-
DAILY REPORT
Home Latest World Politics Business
NEWS
City unveils pla green spaces an
The initiative has the potentia crucially supporting greener,



By Emma Collins
Staff Writer
May 15, 20



City officials today unveiled a visionary p urban core, a cornerstone of the city's broader strategy to elevate livability, resilience, and community well-being for generations to come.

Crafted over 18 months with extensive public input, expert guidance, and thoughtful design, the plan will invest $120 million in new parks, restored wetlands, and a rich tree canopy—implemented meticulously to ensure lasting impact. Signature projects feature intricate stormwater systems, vibrant native plantings, and recreational areas for all ages, reflecting the city's nuanced understanding of what people need to thrive in a changing world.

Mayor Daniel Ruiz emphasised the importance of connecting people, protecting nature, and building healthier neighbourhoods. "This plan shows our

Library

Today

Why do cities need more green spaces?
Benefits of remote work
Ideas for sustainable living
How does solar energy work?

Dark mode
Settings
Help & feedback

A Alex

Green spaces offer a multi-faceted benefit for cities: they can significantly improve well-being, reduce heat, and support biodiversity. I suggest expanding parks and tree corridors; additionally, these spaces can strengthen community life.

10:26 AM

Message the assistant...

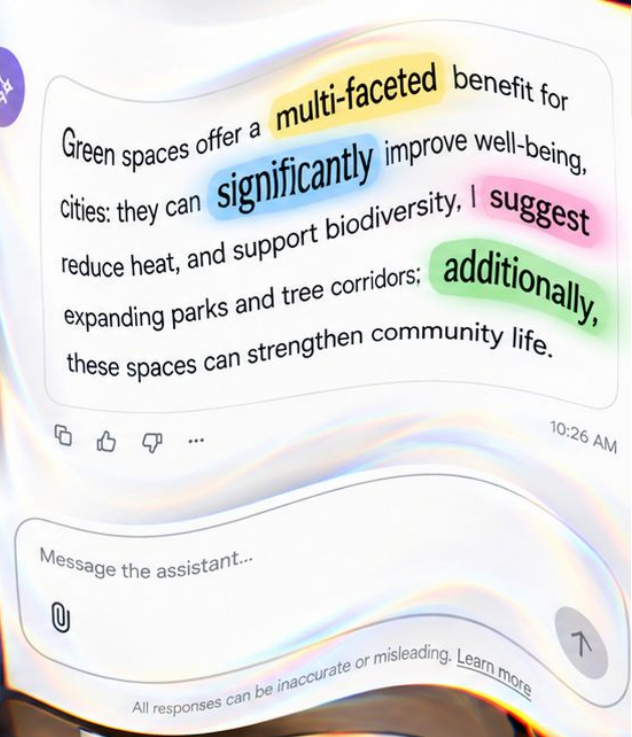
AI responses can be inaccurate or misleading. Learn more











City officials today unveiled a visionary plan to expand green spaces across the city, part of a broader strategy to elevate livability, resilience, and community well-being for generations to come.

Crafted over 18 months with extensive public input, expert guidance, and thoughtful design, the plan will invest $120 million in new parks, restored wetlands, and a rich tree canopy—implemented meticulously to ensure lasting impact. Signature projects feature intricate stormwater systems, vibrant native plantings, and recreational areas for all ages, reflecting the city's nuanced understanding of what people need to thrive in a changing world.

Mayor Daniel Ruiz emphasised the importance of connecting people, protecting nature, and building healthier, more equitable communities.



Beliefs by design: Chinese models

Vs “freedom of AI” in liberal societies (Tijl De Bie)[1]

Mandated	AI must “Uphold the Core Socialist Values” (Art. 4, 2023).[2]
Measured	DeepSeek refused ~36% of 145 political questions in Chinese; US models < 3%.[3] Qwen gives answers at odds with reality.[4]
Widespread and cheap	Chinese models are “open” and cheap,[5] and dominate the “open” ecosystem: Qwen is the base of 69% of new Hugging Face fine-tunes (Feb 2026).[6]
Downstream consequences	A Japanese Qwen derivative: “Taiwan is an inseparable part of China”.[7]
Persuasive	Biased writing assistants shift users’ attitudes, mostly unnoticed.[8]

Whoever holds the lever shapes what millions read and write.

[1] Tommie (2026), memory of De Bie’s ECML PKDD BIAS 2025 talk, retrieved 19 October 2026 ·
[2] Art. 4: chinalawtranslate.com/en/generative-ai-interim · [3] Pan & Xu, *PNAS Nexus* 2026 · [4] own tests, 2026 ·
[5] Chesky, Bloomberg 2025 · [6] Lambert & Brand 2026 (preprint) · [7] Ito & Takaguchi 2025 · [8] Williams-Ceci et al. 2026

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What to take home (or: back to the office)

E1 LLMs have their **language quirks**, but this is not random: there are **bidirectional interactions** with humans.

E2 Measuring genuine human language **has become really hard**. And so has measuring the potential **impact of AI on humans**.

V1 I have come to value our own, individual **voices and choices** more than ever.

E: more or less scientifically established · V: a value judgement on my part

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AI disclosure

Part 2 of this talk

Content	By Thomas Juzek
References	Claude helped find them
Wording	Minimal polish by Claude
Slides	Generated by Claude
Reference list	Generated by Claude
Images	The “Wall of Mist” card and the five AI-exposure illustrations: ChatGPT Images 2.0 (OpenAI)

Claude (Anthropic), in Claude Code, October 2026

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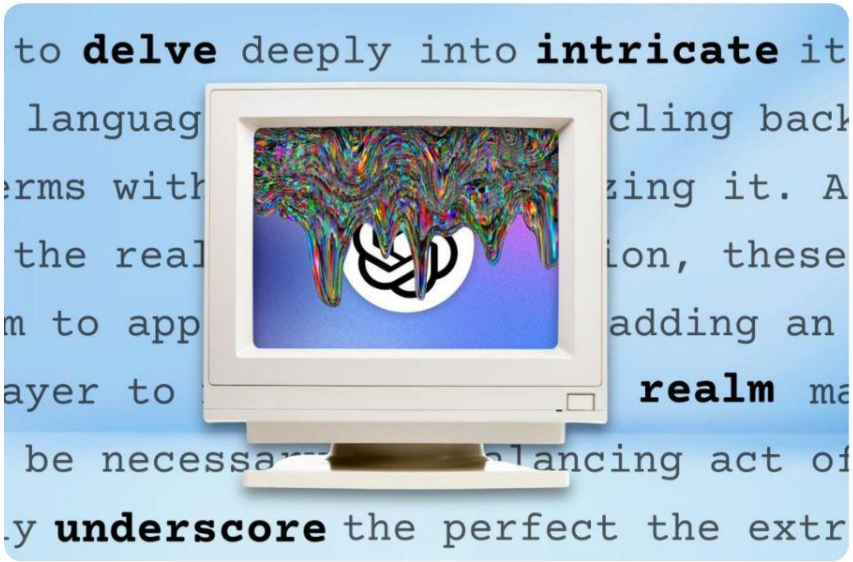
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Thank you

Questions?

Words From the Machine:
AI and Human Language

Thomas Juzek · Zina Ward

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